Understanding Automated Toxicity Detection: Insights from the HateXplain Dataset

Written Report



University of Massachusetts Amherst
Data Analytics and Computational Social Science (DACSS)
Class: 690s Social Media Analysis
By: Aiyman Akbar
Word count: 1801
Date: 12/13/2025

1) Introduction

In recent years, social media has made it much harder to keep harmful and toxic content under control. Insults, threats, slurs, and targeted harassment can spread very quickly, and there is simply too much content for human moderators to review on their own. Because of this, many platforms now rely on machine learning systems to help flag and classify toxic behavior. Even though this is the case, most people do not really know how these systems work or how they learn to make these decisions.

In this write report I talk about some of the key parts of my visualization, titled “Understanding Automated Toxicity Detection: Insights from the HateXplain Dataset”. I will talk detail the data set, visualization approach, audience considerations, and reflections on the storytelling process

In my presentation, I rely on simple data visualizations, Sora AI generated videos, and a structured Prezi layout that walks through the pipeline step by step. Rather than focusing on the math or theory behind machine learning, I tried to focus on giving an intuitive sense of how these systems work in practice and why they matter for online safety. My goal is to make automated toxicity detection easier to understand so that people who are new to this topic can feel more comfortable with how these systems actually work.

2) The Data Source - HateXplain

2.1) Dataset Overview

For this project I chose the HateXplain dataset, a publicly available collection of social media posts that have been manually reviewed by human annotators. Each entry in the dataset contains three parts. The first is the original text of the post, which is broken into individual words, also known as tokens. The second are labels that classify the post as normal, offensive, or hate speech. These labels are assigned independently by each of the three annotators, meaning there are agreements and disagreements between them. The third component is the rationale. Rationales identify the specific words that annotators believed contributed to the offensive or “toxic” meaning of the post. Each annotator provides a separate rationale map, which is visually displayed as an array of ones and zeros, one being toxic and zero being non-toxic for each token in the entry. This allows the dataset to capture subtleties in how different individuals interpret the same text, part of the reason why it is such a valuable dataset.

The dataset also contains information about the target of the harmful content when the annotators determined that a post was directed at a particular group. These target annotations include categories such as race, gender, religion, and other social identities. By combining labels, rationales, and target groups, HateXplain offers a detailed but also transparent record of how people understand toxic language online.

HateXplain includes more than twenty thousand posts, providing a large and diverse sample of real online behavior. The posts vary in length, tone, and context, giving a broad representation of the words and phrases that appear in “toxic” content. The dataset supports both statistical analysis and visualization, making it easy to work with.

2.2) Why This Dataset Is Well Suited for This Project

HateXplain is well suited for this project for a few reasons. First is because it makes the annotation process visible. Many datasets like HateXplain only provide a label for each post, which hides the reasoning behind the decision. HateXplain includes both the labels and the highlighted words that influenced them, allowing viewers to see exactly how humans identify harmful language. This transparency makes it easy for beginners or people less experienced with this kind of data to see a good example.

The dataset also works well for visualization. Patterns such as targeted groups and sentiment differences can be shown clearly through simple charts, some of which I included in my visualization. These visuals help beginners understand how harmful content appears across many posts and why human annotations are important.

The last one I will talk about is how well this dataset fits the story narrative I was trying to create in my presentation. It allowed me to move from raw text, to annotation, to processing, and finally to model behavior, all while using concrete examples and being able to tie them back to the dataset. I feel this helped demystify the process for viewers who haven't learnt about it before, and shows how toxic content becomes data that a model can learn from.

3) Visualization approach and Audience Considerations.

The visual design of this project was shaped a lot by the goal of making the process of toxicity detection understandable to viewers who are new to machine learning. That is why I combined these two sections and decided to talk about both together. Because the topic I chose involves several abstract steps, I used visual storytelling to break the process into clear and intuitive stages. This approach helped me decide on the structure of the presentation and the style of every chart and animation.

By combining real dataset examples, simple language and a good presentation aesthetic I tried to design the presentation to give beginners an approachable and engaging look at how toxic content is identified, annotated, processed, and classified.

3.1) Visualization Approach

Prezi served as the main platform I used because it is interactive and allows me to move through space in between slides which not only looks cool but also keeps whoever is watching engaged. I tried to make it so viewers follow a guided path that mirrors the steps in the toxicity detection pipeline, starting with online posts and moving through annotation, data processing, supervised learning, and model output. I wanted each stage to build on top of each other in a way where the viewers would actually understand and be interested throughout. Something I also tried to include in the slides with words explaining topics were related pictures, just to make them look less daunting and hopefully help people understand better.

That same reason is why I chose to include the Sora AI generated videos, and specifically in the locations that I did. I wanted these generative videos to serve as sort of a short break in between the word

based slides to reinforce the things I talked about before and show it in a different way for people to understand. I am a visual learner and have always learnt better through visuals rather than text alone, and that influenced my choice to include those videos also. The videos show simplified versions of concepts I was trying to convey in my story in an abstract way.

The data visualizations that I generated are meant to highlight patterns in the HateXplain dataset. Each chart communicates a single insight and the first four are quite simple. I chose to include the sentiment analysis box-and-whisker plots at the end party because I thought it was an interesting visualization, but also to show beginners a bit or a more complex kind of visualization to show all the variety out there in what you can do with this dataset.

I chose to include things like the screenshots of the structure of an entry in the dataset with the audience in mind. I wanted there to be an example of what I was talking about actually looked liked, such as the annotations and rationales. I also wanted to show people who were curious or have never worked with this kind of data what an entry would look like, its structure and what it all means.

3.2) Audience Considerations

Even though nearly all of my visual choices were shaped by the audience considerations, not all the audience considerations were visual. I tried to keep the language simple and conversational, and technical terms were introduced only when necessary and not too often. Concepts like rationales and supervised learning I tried to explain through or tie back the dataset so that viewers could connect new ideas to concrete illustrations, not just words.

I tried to make it so each slide focused on a single idea to guide the audience through the material in a manageable way. The pacing and structure were also shaped by considering the audience, I tried to keep the pace at which concepts were introduced steady and in a logical order that mirrors the steps of the toxicity detection pipeline.

4) Reflections on the Storytelling Process

Creating this project was mostly about turning a complicated technical process into something that beginners could actually follow. One thing that worked well was structuring the story as a clear pipeline, starting from raw toxic posts and ending with model predictions. I think that structure makes it easier for viewers to understand how each step connected to the next without needing deep technical knowledge. Using real examples from the HateXplain dataset also helped a lot. I think it helps people see the actual posts, labels, and explanations that shaped the data, which makes the process feel more real and grounded rather than abstract, which is what I was going for.

There were also ideas I wanted to explore but could not fully implement. One example was creating a more character driven narrative that followed a single post through every stage of the pipeline. That would have made the process even more intuitive, but I wasn't sure how to go about it effectively.

Looking back, there are a few things I would improve. I would add more interactive elements so viewers could experiment with the data or test simple examples themselves. I would also focus more on model mistakes and include clearer side by side comparisons between human judgments and model predictions. I think this would help reinforce the idea that automated moderation systems are useful but not perfect.

Overall, telling the story I tried to through my visualization was a really great learning experience. It made me think not just about building models, but about how to explain them in a way that is clear and engaging. I thought this part would be a lot easier than it actually was, because I was thinking if I understand it then surely I can explain it simply.

5) Conclusion

This project gave me the opportunity to think carefully about how complex technical systems can be communicated to a nontechnical audience, something I haven't done before. By working with the HateXplain dataset, I was able to explore how toxic content is labeled, structured, and used in machine learning, while also considering how the whole process can be explained clearly through storytelling and visualization.

After completing this project and report, the experience highlighted the importance of combining data analysis with thoughtful presentation. It helped me understand truly that machine learning is just as much about models and results as it is also about how those ideas are communicated.

6) Links:

- Link to HateXplain dataset: HateXplain
- Link to my Prezi: Presentation
- Link to my Python code: Data Visualization